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Energy materials

Porous carbon derived from Congo Red-adsorbed Chitosan for supercapacitor electrodes

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ABSTRACT

Reuse of dye-containing wastewater for valuable products offers significant potential. Chitosan, a natural high-molecular-weight porous material, served as an effective carrier for dye molecules. This study uses chitosan to adsorb Congo Red, preparing hierarchical porous carbon materials, which are then applied in supercapacitors. The findings reveal that chitosan achieves an impressive adsorption capacity of 1450 mg g^{-1} for Congo Red. When the chitosan–Congo Red composite material-based carbon is used as the electrode material, it shows an outstanding specific capacitance of 318 F g^{-1} at a current density of 1 A g^{-1}. Furthermore, due to its high content of heteroatoms such as N, O, and S, the double-layer supercapacitor fabricated with this electrode material demonstrates a specific capacitance of 149.49 F g^{-1} at 1 A g^{-1}, with an energy density of 8.77 Wh kg^{-1} at a power density of 335.35 W kg^{-1}. These results highlight the considerable potential of this material after adsorption of dye-containing wastewater for applications in the energy storage field.

Introduction

Dye-containing wastewater is one of the most critical environmental challenges facing humanity. Each year, up to 200,000 tons of dyes containing toxic, carcinogenic, and mutagenic substances are released into wastewater [1], causing severe environmental damage. Hence, wastewater containing dyes should be efficiently treated with environmentally friendly methods to prevent adverse impacts on the environment, public health, and natural water supplies [2]. Adsorption is a particularly effective, efficient, and cost-effective method for treating dye-containing wastewater [3–5]. Chitosan, zeolite, montmorillonite, and activated carbon, are common adsorbents for treating dyes such as Congo Red in wastewater [6–9]. Shivani [10] developed a novel Chitosan–lignin composite in a 1:1 ratio for removing MB from simulated dye wastewater. Other approaches include cross-linked Chitosan [11], chitosan composite hydrogel spheres [12], and chitosan–iron oxide

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nanocomposite membranes [13]. However, many chitosan-based materials are often discharged into the environment or incinerated after treating Congo Red wastewater, resulting in secondary pollution and resource waste. In this study, we investigated the recovery of chitosan after adsorption and its application in the manufacture of supercapacitors.

Supercapacitors are becoming a new energy storage solution, merging features of both conventional capacitors and batteries [14]. Chitosan has garnered significant interest from researchers in the development of diverse electrode materials with excellent electrical properties for supercapacitors. Chitosan-based electrodes for supercapacitors can be classified into chitosan-based composite electrodes and chitosan-based porous carbon electrodes. For instance, chitosan/graphene/polyaniline ternary nanocomposite electrodes exhibit a peak specific capacitance of 1519 F g^{-1} and a conductivity of 2.9745 S cm^{-1}, outperforming all currently available electrode materials [15]. However, the material's cycle stability and cycle longevity are shorter than those of carbon-based materials used for comparable uses. Over the last few years, chitosan-derived porous carbon electrodes with heteroatoms such as N, P, B, and O have been intensively studied to improve the various properties of supercapacitor. These findings provide compelling evidence of potential utilization of chitosan-based materials in electrode manufacturing for supercapacitors.

In this work, chitosan was firstly used to adsorb Congo Red, then prepared hierarchical porous carbon materials after high-temperature carbonization, and applied in supercapacitors. The chitosan–Congo Red (CS/CR) composite material-based carbon electrode employed in supercapacitors shows an impressive specific capacitance of 318 F g^{-1} at a current density of 1 A g^{-1}. Additionally, due to its high levels of heteroatoms such as N, O, and S, the assembled supercapacitor demonstrates a specific capacitance of 149.49 F g^{-1} at 1 A g^{-1}, with an energy density of 8.77 Wh kg^{-1} at a power density of 335.35 W·kg^{-1}. Therefore, this study provides an economical and efficient method for recycling chitosan-based post-adsorption materials, which can be utilized as a carbon source for developing high-performance supercapacitor electrode materials.

Experimental section

Materials

Chitosan and Congo Red were obtained from Aladdin Reagent Co., Ltd. Hydrochloric acid and anhydrous ethanol were purchased from Beijing Chemical Factory. Conductive acetylene black came from Beijing Zhuangshi Technology Co., Ltd. Polytetrafluoroethylene (60% solid content) was sourced from Cyber Electrochemical Materials on Taobao. Potassium hydroxide was acquired from Tianjin Guangfu Technology Development Co., Ltd., and high-purity nitrogen (99.999%) was supplied by Beijing Huanyu Jinghui Jincheng Gas Co., Ltd. All chemicals were of analytical grade and used without further purification.

Chitosan adsorption of Congo Red

Congo Red solutions with varying concentrations were prepared and their absorbance at $\lambda = 498$ nm was measured to create a concentration–absorbance standard curve. As illustrated in Fig. 1, the fitted standard curve equation is ABS = 0.00414*C*−0.00014. In subsequent experiments, the absorbance of the Congo Red solution was measured post-adsorption, compared it to this standard curve, and the amount of adsorption was calculated.

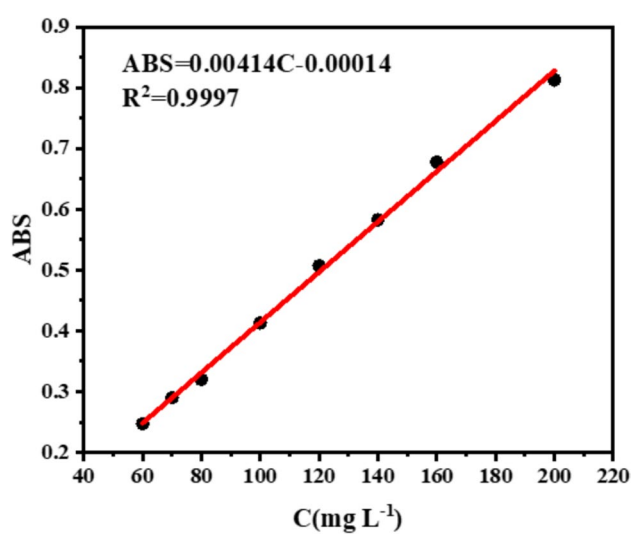


Figure 1 UV absorption spectra standard curve for Congo Red solutions.

Effect of adsorption time

1.5 g Congo Red is added into 100 mL water to prepare a saturated Congo Red solution. Dilute hydrochloric acid is used to adjust the pH to acidic. Next, 1.0 g chitosan powder is added to the above solution. The mixture was placed in a constant-temperature water bath at 25 °C and stirred continuously. Samples were taken from the solution, centrifuged to separate the supernatant, and its absorbance was measured each hour. The above process repeated until the concentration of the supernatant stabilized, indicating that adsorption equilibrium had been reached.

The effect of Congo Red concentration

Congo Red solutions with concentrations of 30, 50, 80, 100, 300, 500, 800, 1000, and 1500 mg L^{-1} were prepared. The pH of the solution is adjusted to 5 with dilute hydrochloric acid. 1.0 g of chitosan is added into the above different concentrations of Congo Red solution, and then shaken in a constant-temperature water bath at 25 °C for 24 h until the adsorption equilibrium was achieved, then centrifuged to measure the absorbance of the supernatant of the samples to get the value of the concentration of the supernatant.

Preparation of CRK porous carbon

After achieving adsorption equilibrium, the material was separated and dried for 8 h to obtain the fully adsorbed composite carbon precursor CS/CR. The dried CS/CR precursor was added into a tube furnace and heated at a rate of 5 °C per minute to 600 °C, 700 °C, and 800° for 3 h under a nitrogen atmosphere. The samples were repeatedly washed and centrifuged until their pH was neutral. The resulting carbonized samples were labeled CRK600, CRK700, and CRK800, respectively, where C, R, and K stand for chitosan, Congo red, and carbonization, respectively. The sample obtained from carbonizing pure chitosan at 700 °C was labeled CS700 (Fig. 2).

Adsorption performance test

Decolorization rate R:

$$R = \frac{A_0 - A_t}{2a} \times 100\% \tag{1}$$

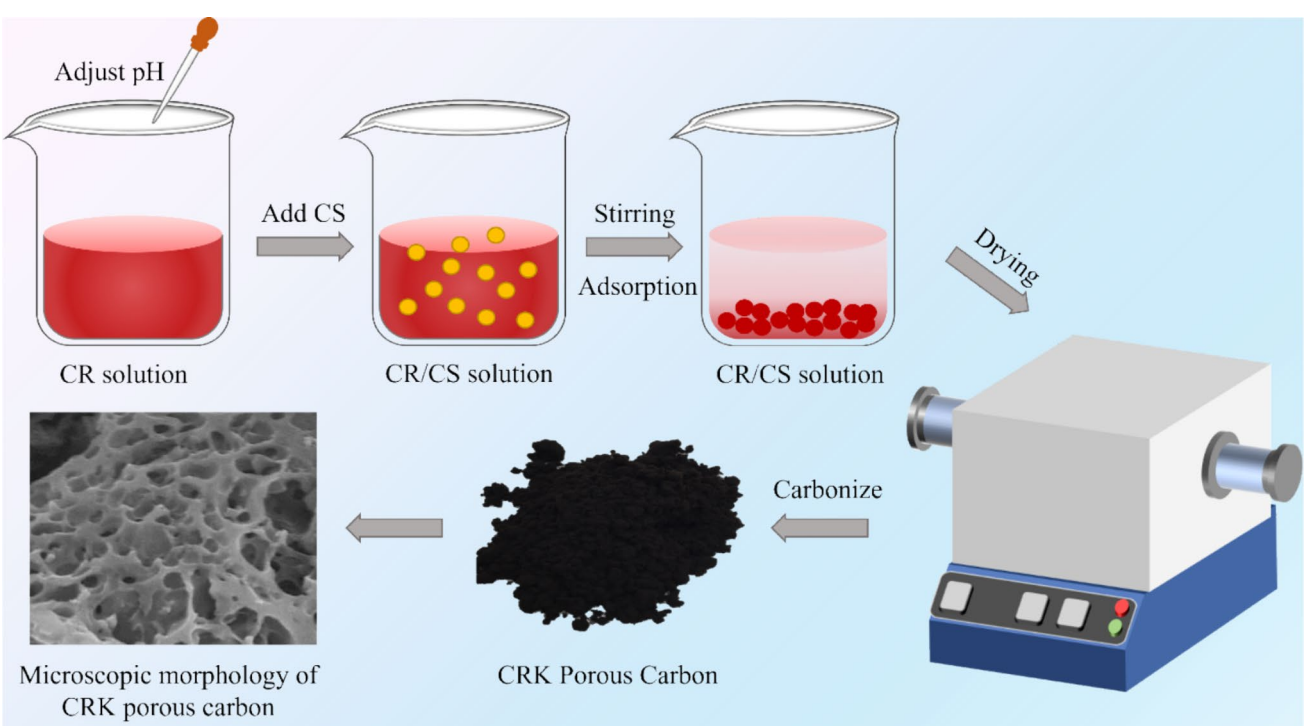


Figure 2 Flow diagram for the synthesis of CRK porous carbon.

A_0 denotes the maximum absorbance of the initial simulated dye-containing wastewater measured at 497 nm, and A_t represents the maximum absorbance of the dye-containing wastewater measured at 498 nm after t hours of treatment.

Adsorption capacity Q_t(mg g^{-1}):

$$Q_t = \frac{C_0 - C_t}{m} \times V \tag{2}$$

C_0 (mg L^{-1}) refers to the starting concentration of the solute in the solution, C_t (mg L^{-1}) indicates the solute concentration after a specific adsorption time t, V (L) represents the volume of the solution, and m(g) denotes the weight of the adsorbent material.

Equilibrium adsorption capacity Q_e(mg g^{-1}):

$$Q_e = \frac{C_0 - C_e}{m} \times V \tag{3}$$

C_0 (mg L^{-1}) represents the initial concentration of the solute in the solution to be adsorbed, C_e (mg L^{-1}) denotes the concentration of the solute at adsorption equilibrium, V (L) is the volume of the solution, and m(g) is the mass of the adsorbent material.

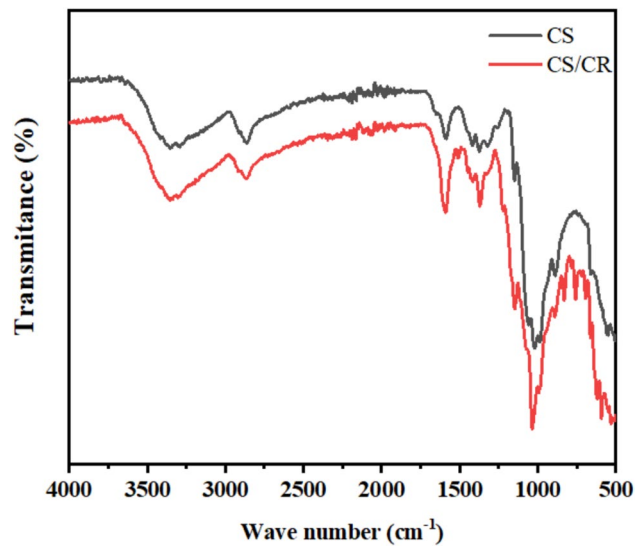


Figure 4 FTIR plots of chitosan before and after adsorption of Congo red.

Fabrication of supercapacitor and electrochemical performance testing

Electrodes were prepared by blading the mixture of CRK, acetylene black, and polytetrafluoroethylene at a ratio of 8:1:1 in ethanol onto nickel foam, then pressing

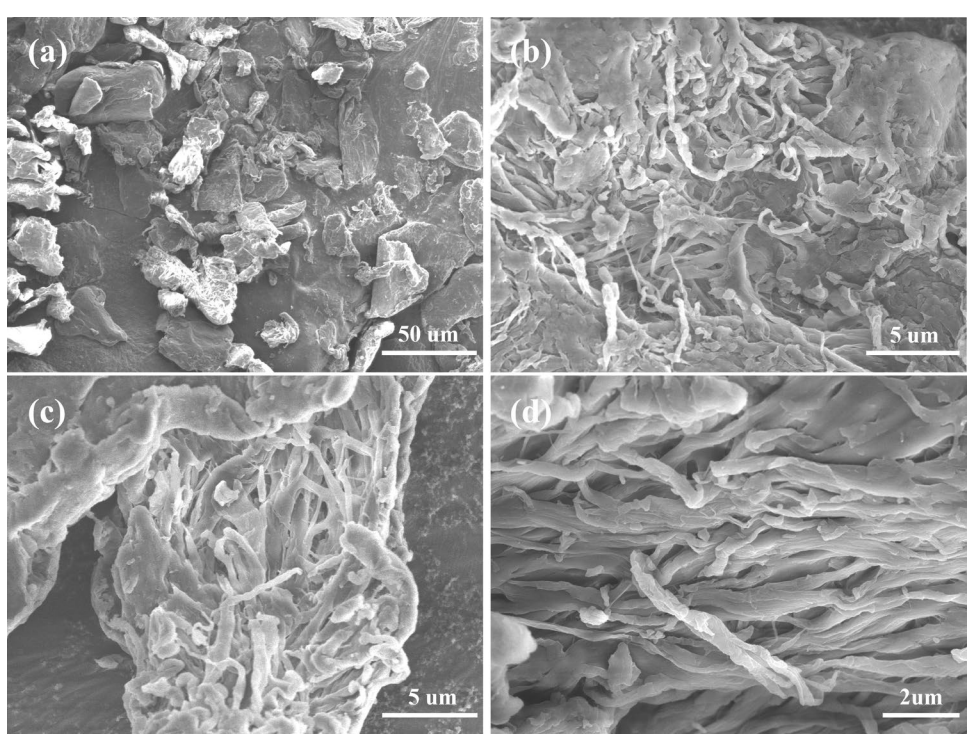


Figure 3 SEM image of CS.

at 10 MPa for approximately 3 s and drying at 80 °C. To assemble the supercapacitors, the electrode sheets were stacked to create a double-layer supercapacitor using a 3-M KOH solution as the electrolyte and fiber paper as the separator, labeled as CRK600//CRK600, CRK700//CRK700, and CRK800//CRK800, respectively. The working area was 1 cm^2, and the areal mass loading of CRK for the electrode was 3 mg cm^{-2}.

The electrochemical properties of the electrode materials were analyzed using a CHI660E electrochemical workstation in a three-electrode system using Pt sheet as the counter electrode, Hg/HgO electrode as the reference electrode, and a 3-mol·L^{-1} KOH solution acted as the electrolyte. Cyclic voltammetry (CV) and constant-current charge/discharge (GCD) techniques were carried out over a voltage range of −1 to 0 V. The frequency response of the electrodes was measured from 0.01 to 100 kHz. Additionally, AC impedance spectroscopy (EIS) was used to investigate the capacitive-resistive properties of the carbon samples within the same frequency range. The mass-specific capacitance(C_s, F g^{-1}) was calculated using the following equation:

$$C_S = \frac{I \times \Delta t}{m \times \Delta V} \tag{4}$$

I(A) denotes the discharge current, Δt(s) represents the discharge time, m(g) is the mass of the active material, andΔV(V) is the potential window.

The electrochemical performance of the supercapacitor was evaluated using a symmetric two-electrode system. GCD and cyclic CV tests were conducted within a voltage window of −1.3 to 0 V. EIS was performed over a frequency range of 0.01 to 100 kHz. The capacitance (C_{cell}, F g^{-1}), energy density (E, Wh kg^{-1}), and power density (P, W kg^{-1}) were calculated using the following formulas:

$$C_{cell} = \frac{2 \times I \times \Delta t}{m \times \Delta V} \tag{5}$$

$$E = \frac{C_{cell} \times \Delta V^2}{8 \times 3.6} \tag{6}$$

$$P = \frac{3600 \times E}{\Delta t} \tag{7}$$

I(A) represents the discharge current, Δt(s) stands for the discharge duration, m(g) refers to the mass of the active material in one electrode, and ΔV(V) denotes the voltage window.

Material characterization

The adsorption capacities of CRK were measured using a UV–Vis spectrophotometer (Model: UV-6100PC). The specific surface area and pore size distribution of CRK were determined through nitrogen adsorption–desorption experiments (Mike ASAP2460), with detailed analysis conducted using Brunauer–Emmett–Teller (BET) and Density Functional Theory (DFT). The microscopic structure, surface morphology, and elemental distribution of CRK were examined using Field Emission Scanning Electron Microscopy (Hitachi S4800), Transmission Electron Microscopy (JEM 1200EX), and Energy-Dispersive

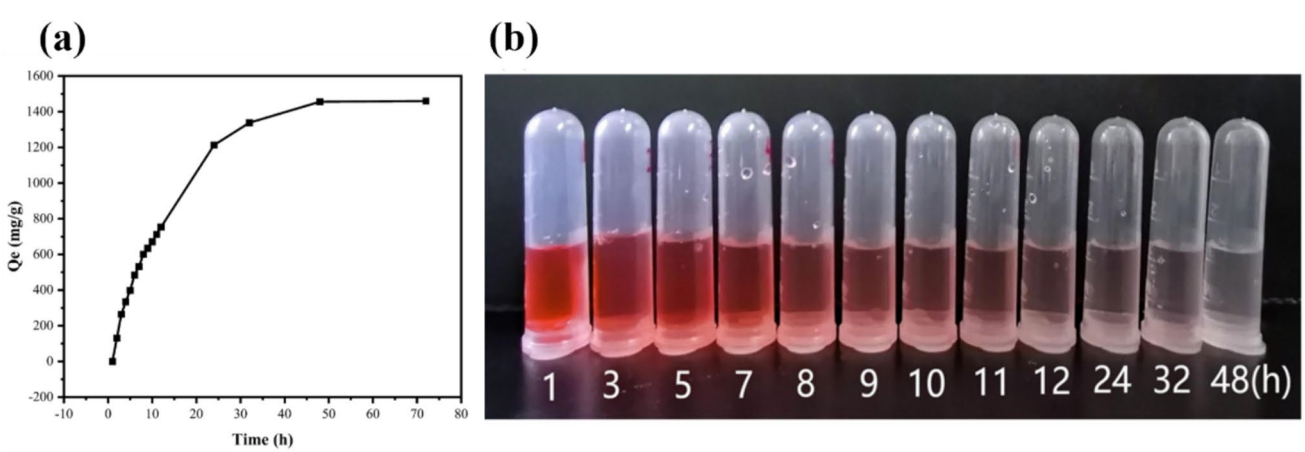


Figure 5 Variation of Congo Red adsorption (**a**) and color (**b**) with time.

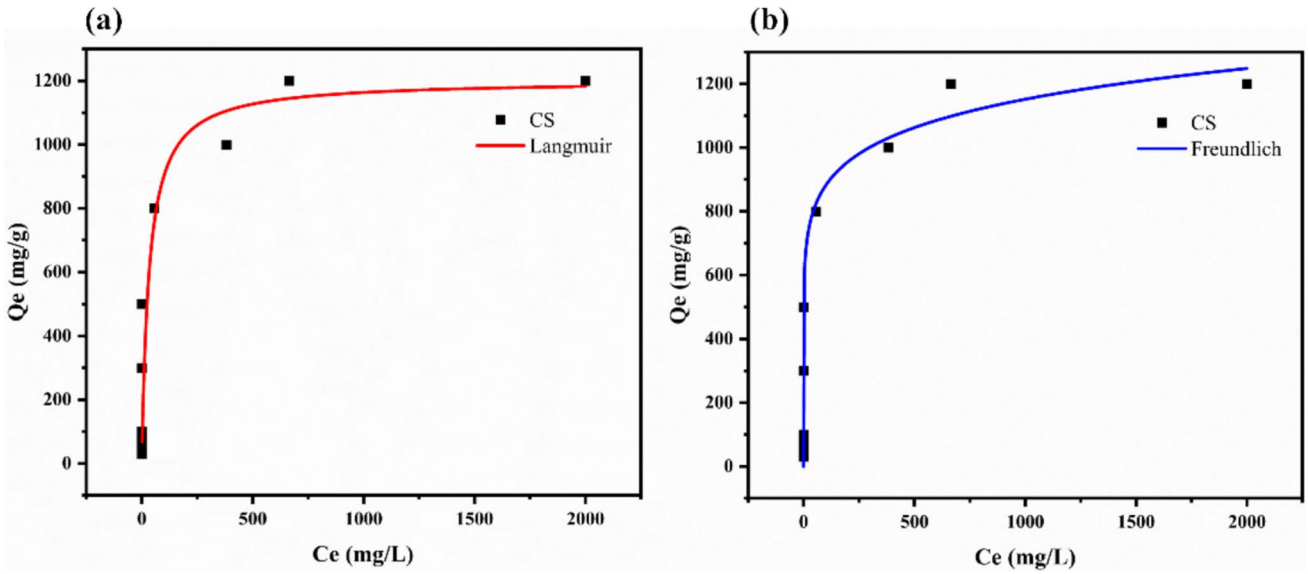


Figure 6 Congo Red adsorption model fit **a** Langmuir **b** Freundlich.

Table 1 Isothermal adsorption fitting parameters of the samples

Langmuir mold			Freundlich mold		
Q_m (mg g^{-1})	K_L (10^{-2}L mg^{-1})	R^2 adj	K_F (mg g^{-1})	n	R^2 adj
1202.39	0.03	0.85	515.86	8.60	0.80

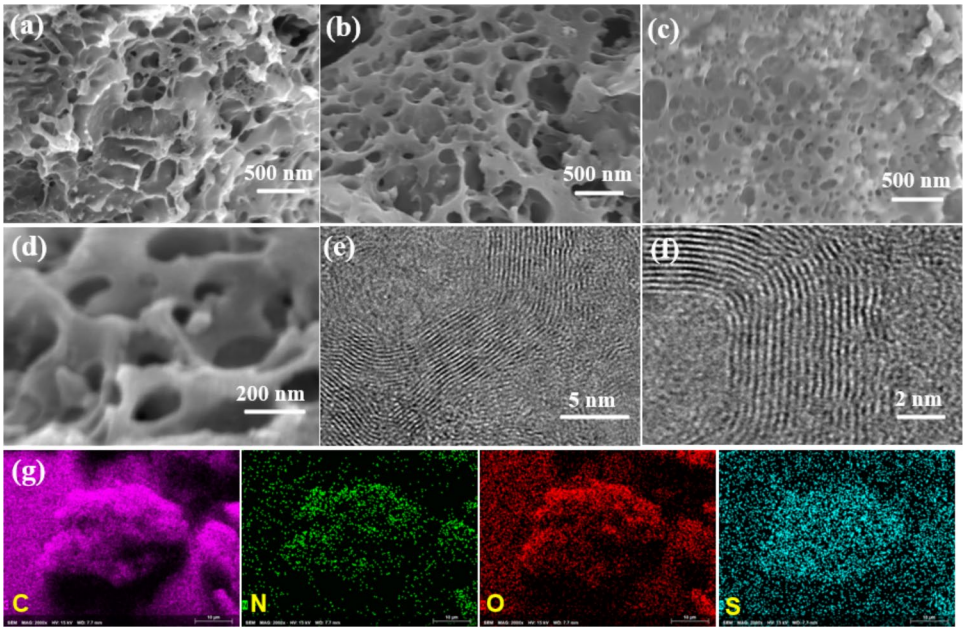


Figure 7 **a** SEM image of CRK600, **b** SEM image of CRK700, **c** SEM image of CRK800, **d** SEM image of CRK700, **e**, **f** TEM image of CRK700, and **g** EDX mapping of C, N, O, and S elements of CRK700.

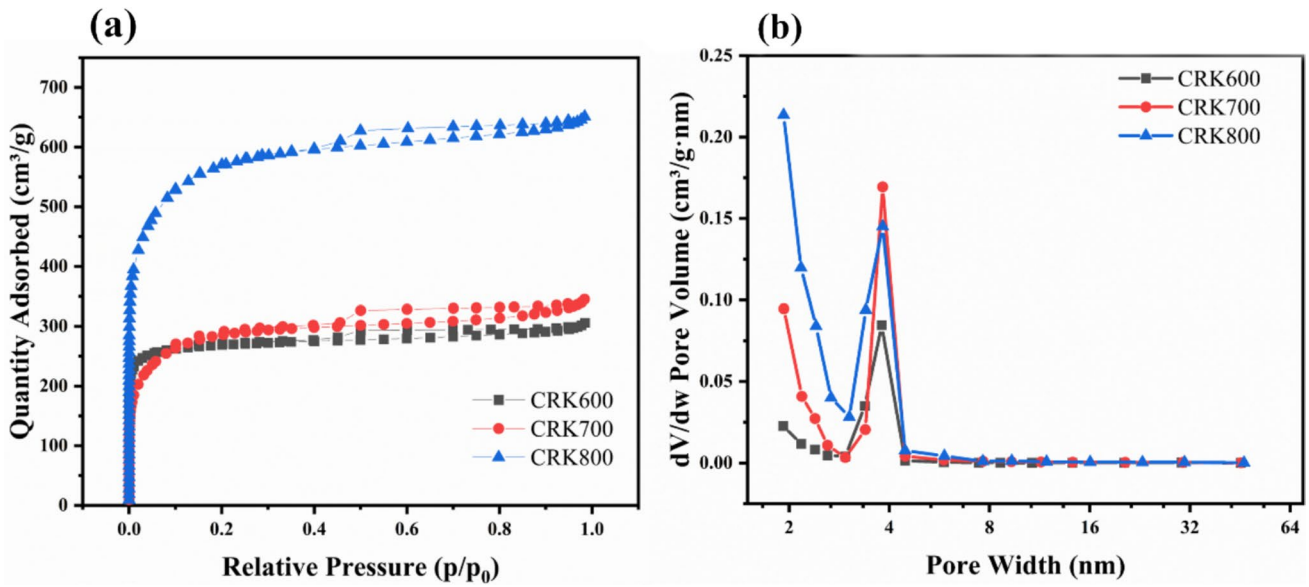


Figure 8 **a** N_2 adsorption/desorption curves, **b** pore size distribution of CRK samples.

X-ray Spectroscopy (EDX). The molecular and crystalline structures were identified by X-ray Diffraction (Brook D8 Advance, Cu-Kα) and Raman Spectroscopy (Renishaw inVia, 532 nm). X-ray Photoelectron Spectroscopy (XPS, Thermo ScaLab 250Xi) was employed to analyze the types, quantities, oxidation states, and bonding states of the atoms in CRK.

Results and discussion

SEM was used to characterize the microstructure of chitosan. Figure 3a shows that chitosan has a blocky structure with a rough surface, approximately 50 µm in size. Figures 3b–d reveal that chitosan exhibits a fibrous structure, displaying an open network composed of cross-linked nanofibers with nanoscale pores between them. These interstitial spaces facilitate mass diffusion, indicating that Congo Red dye molecules may permeate into this structure.

The FTIR spectra of chitosan and chitosan/Congo Red are shown in Fig. 4. In the FTIR spectrum of chitosan/Congo Red, the peaks at 1564 cm^{-1} and 1320 cm^{-1} become more intense after the adsorption of Congo Red. These peaks correspond to the stretching vibrations of the azo bonds and C–C bonds in Congo Red [16]. The characteristic peaks of chitosan at 3443 cm^{-1}, 2932 cm^{-1}, 1655 cm^{-1}, 1564 cm^{-1}, and 1387 cm^{-1} are assigned to the stretching vibrations of -OH and -NH_2 groups, C-H stretching vibrations, the amide I band which includes N–H bending vibrations, the amide II band involving N–H stretching vibrations, and the symmetric bending of -CH_3 groups, respectively [17]. In the FTIR spectrum of chitosan/Congo Red, the peak at 1655 cm^{-1} (amide II band, associated with N–H bending vibrations) is nearly absent, indicating that the -NH_2 groups play a significant role in the adsorption of Congo Red [16]. Furthermore, no peaks corresponding to sulfonic acid groups are observed in the 1200–1000 cm^{-1}

Table 2 Pore structure parameters of CRK samples

Sample	S_{BET} ($m^2\ g^{-1}$)	V_{total} ($cm^3\ g^{-1}$)	V_{micro} ($cm^3\ g^{-1}$)	V_{meso} ($cm^3\ g^{-1}$)	D (nm)	V_{Meso}/V_{total}(%)
CRK600	1067.80	0.474310	0.401543	0.072767	1.7768	15.34
CRK700	1089.91	0.534719	0.425591	0.109128	1.9624	20.41
CRK800	2152.12	1.009166	0.879785	0.129381	1.8757	12.81

*S_{BET} refers to the overall BET surface area; V_{total} denotes the total pore volume; V_{micro} indicates the micropore volume; V_{meso} signifies the mesopore volume; D represents the mean pore diameter; V_{Meso}/V_{total} corresponds to the ratio of mesopores

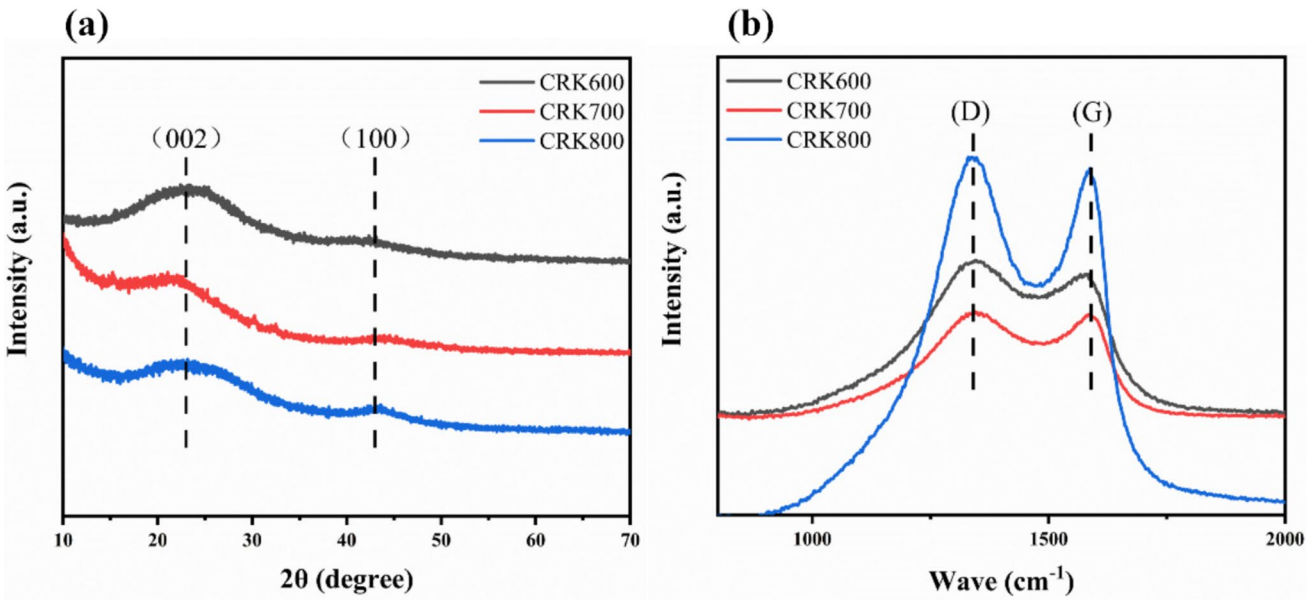


Figure 9 **a** XRD graph of CRK sample, **b** Raman spectrum of CRK sample.

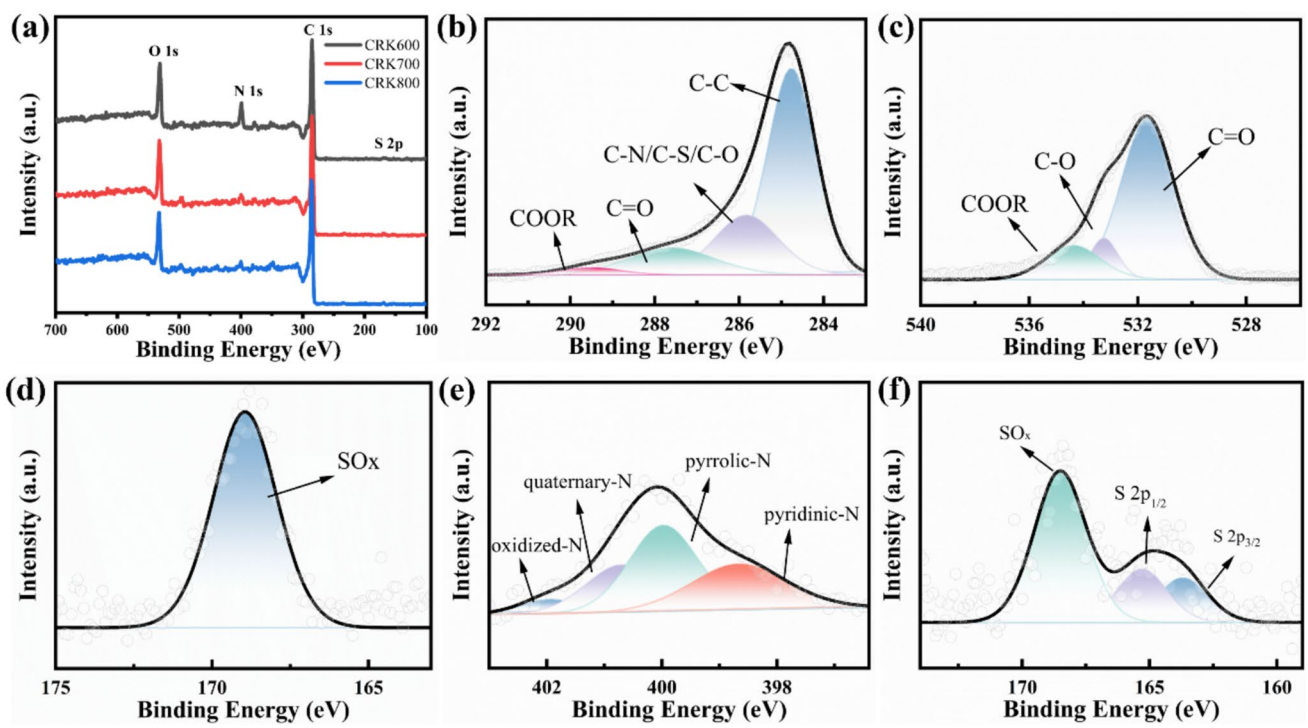


Figure 10 **a** XPS full spectrum of CRK sample, **b** C 1 s of CRK700, **c** O 1 s of CRK700, **d** S 2p of CRK700, **e** N 1 s of CRK700, and **f** S 2p high-resolution spectrum of CRK600.

range, suggesting that the sulfonic acid groups of Congo Red are involved in the adsorption process.

The adsorption experiments detailed in Sect. 2.2.1 examined the equilibrium time and capacity of chitosan for Congo Red. Figure 5a indicates that equilibrium is reached after two days at room temperature,

Table 3 Content of each atom in CRK samples

Samples	C 1s (at.%)	N 1s (at.%)	O 1s (at.%)	S 2p (at.%)
CRK600	75.09	9.27	15.05	0.59
CRK700	78.81	4.9	15.42	0.88
CRK800	81.84	3.17	14.35	0.64

and the equilibrium adsorption amount is calculated to be 1450 mg g^{-1} by Eq. 3. Figure 5b further shows that after 48 h, the red color in the supernatant has nearly vanished, confirming that Congo Red is almost entirely adsorbed by chitosan at a concentration of 15,000 mg L^{-1}.The mechanism of adsorption of Congo Red by chitosan is further explored by changing the initial concentration of Congo Red solution. The experimental results are fitted with Langmuir and Freundlich equations, as expressed in Eqs. 8 and 9 respectively.

$$Q_e = Q_m \frac{K_L C_e}{1 + K_L C_e} \tag{8}$$

$$Q_e = K_F C_e^{\frac{1}{n}} \tag{9}$$

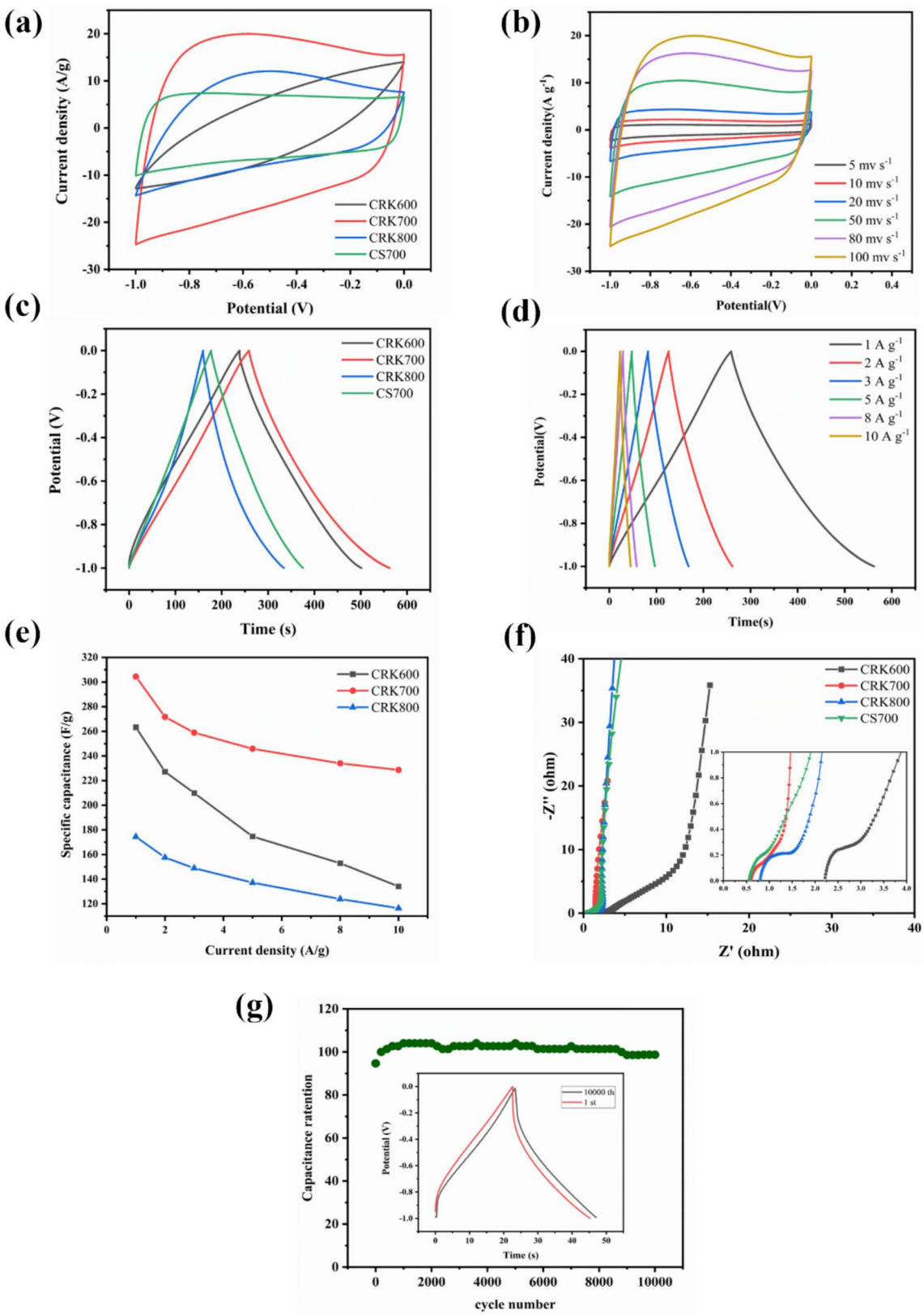
In the equations, Q_m and Q_e denote the maximum adsorption capacity and the equilibrium adsorption capacity, respectively. K_L and K_F represent the Langmuir constant and the Freundlich constant, respectively. C_e is the equilibrium concentration, and n is the empirical constant. As shown in the adsorption model fitting curves in Fig. 6, the Langmuir equation provides a better fit (higher R^2 value) than the Freundlich equation, indicating that the adsorption of Congo Red by chitosan is more consistent with the Langmuir adsorption model. This suggests that during the adsorption of Congo Red onto chitosan, specific adsorption sites on the chitosan surface interact with the Congo Red molecules, leading to their binding to the chitosan surface (Table 1).

Analysis of physical and chemical properties of porous carbon

Figure 7a–d shows that samples carbonized at different temperatures exhibit highly dense porous structures and interconnected networks, which facilitate rapid ion transport. Figure 7d indicates that the pore sizes are approximately 100–200 nm. The presence of mesopores and macropores is advantageous for electrolyte storage. Additionally, the extensive porous structure provides a large specific surface area, enhancing hydrophilicity and reducing diffusion resistance in aqueous electrolytes [18]. Figure 7e–f displays the electron microscope images of CRK700, in which lattice fringes are distinctly visible, with their distance measured at 0.22 nm, corresponding to the (100) crystalline plane of graphitic carbon [19]. This indicates the formation of minute graphite microcrystals within the amorphous carbon matrix. However, these microcrystals are small in size and limited in quantity. The EDX mapping image of CRK700 in Fig. 7g demonstrates an even distribution of carbon (C), nitrogen (N), oxygen (O), and sulfur (S), confirming the successful doping of heteroatoms into the carbon framework following carbonization.

Figure 8a shows that all carbon materials exhibit type-IV shape. A steep slope at $p/p_0 < 0.1$ indicates a substantial presence of micropores. The hysteresis loop observed between $p/p_0 = 0.4$–1.0 signifies the presence of mesopores [20]. The upward curvature of the isotherm at p/p_0 close to 1.0, along with increased adsorption, suggests the existence of macropores, which facilitate electrolyte transport and charge transfer during the charging and discharging processes [21]. Figure 8b also shows that all the samples possess a rich presence of microporous, mesoporous, and microporous structures. As shown in Table 2, the SSAs of CRK600, CRK700, and CRK800 are 1067.80 m^2 g^{-1}, 1089.91 m^2 g^{-1}, and 2152.12 m^2 g^{-1}, respectively.

The XRD pattern (Fig. 9a) reveals two broad peaks centered at $2\theta = 24.1°$ and 43.7°, corresponding to the (002) and (100) reflections of graphite. This indicates a relatively low degree of graphitization [22], which is consistent with the TEM images. In the Raman spectra (Fig. 9b), all samples exhibit characteristic peaks for carbon materials at 1325 cm^{-1} and 1582 cm^{-1}, which correspond to the D band (defects or disorder) and G band (graphitic carbon structure or order), respectively. The ID/IG ratio of CRK700 (1.01) is marginally lower than that of CRK600 (1.06), suggesting a modest enhancement in the short-range structural ordering of the carbon framework or a slight decrease in defect density upon reaching 700 °C. This structural change contributes to enhancing the conductivity of the carbon skeleton [23]. It should be noted, however, that all ID/IG values remain close to 1.0, which collectively indicates a predominantly amorphous carbon structure with a high degree of disorder across


(a)
Current density (A/g)
Potential (V)
CRK600
CRK700
CRK800
CS700
(b)
Current denity(A g-1)
Potential(V)
5 mv s-1
10 mv s-1
20 mv s-1
50 mv s-1
80 mv s-1
100 mv s-1
(c)
Potential (V)
Time (s)
CRK600
CRK700
CRK800
CS700
(d)
Potential(V)
Time(s)
1 A g-1
2 A g-1
3 A g-1
5 A g-1
8 A g-1
10 A g-1
(e)
Specific capacitance (F/g)
Current density (A/g)
CRK600
CRK700
CRK800
(f)
-Z'' (ohm)
Z' (ohm)
CRK600
CRK700
CRK800
CS700
(g)
Capacitance ratention
cycle number
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1 st
Potential (V)
Time (s)

◂**Figure 11** **a** Cyclic voltammetry curves of carbon samples at different temperatures of 100 mV s^{-1}, **b** cyclic voltammetry curves of CRK700 at different scanning speeds in a three-electrode system, **c** GCD curves of each sample at 1 A g^{-1}, **d** GCD curves of CRK700 at various current densities, **e** comparison of the capacitance stability of each carbon sample at different current densities, the **f** Nyquist curves of each carbon sample (inset: amplification curves in the high-frequency area), **g** cycling performance of CPK700 at 10 A g^{-1} (inset: GCD curves at the beginning of the cycle and after the cycle).

all samples. The slight increase in the ID/IG ratio for CRK800 (1.03) may be attributed to the formation of additional defects resulting from heteroatom removal at higher carbonization temperatures.

The elemental composition and chemical valence of the materials are further analyzed by XPS. The CRK-x sample displays a clear O 1 s peak, an N 1 s peak, and a faint S 1 s peak, along with the primary C 1 s peak in Fig. 10a. Table 3 lists the specifics of each element. Notably, the carbon content increases with higher carbonization temperatures, while the nitrogen content decreases. This decrease is attributed to the gradual volatilization of nitrogen at elevated temperatures [24].

In addition, the presence of functional groups in CRK700 is further investigated. Figure 10b reveals four chemical bonds in CRK700: C–O (284.7 eV), C–X (285.6 eV), C = O (287.7 eV), and COOR (289.8 eV). In Figure 10c, O 1s is deconvoluted into three peaks of C = O (531.4 eV), C–O (532.8 eV), and COOR (534.1 eV). Figure 10e shows the deconvolution of N 1s spectrum separates four distinct peaks at 398.1 eV (pyridinic N), 399.9 eV (pyrrolic N), 401.3 eV (quaternary N), and 402 eV (oxidized N) [25]. Pyrrolic and pyridinic N, which contain unpaired electrons, are more active in enhancing capacitance compared to other nitrogen types [26], while quaternary ammonium salts enhance electron transfer across carbon planes [27, 28]. The high proportion of these nitrogen forms suggests that CRK has a strong potential for excellent energy storage. S 2p spectrum (Fig. 10f) of CRK600 can be resolved into three peaks corresponding to the functional groups at 163.2 eV (S $2p_{3/2}$), 165.8 eV (S $2p_{1/2}$), and 169.0 eV (SO_x). The first two peaks are assigned to sulfur doped into the carbon matrix in the form of C–S–C covalent bonds, which confirm the successful sulfur doping [29, 30]. In addition, the detection of a peak at 169.0 eV indicates the co-existence of a minor fraction of sulfur that was oxidized to higher-valent oxysulfur species. The absence of the C–S–C bond signature in the CRK700 S 2p spectrum, coupled with the sole presence of the SO_x peak, reveals a complete transformation of sulfur chemistry. This suggests that at 700 °C, the doped C–S–C bonds are cleaved and the sulfur is ultimately converted into thermodynamically favored sulfate/sulfonate groups [31]. While this eliminates the pseudocapacitive contribution of sulfur doping, the introduced polar surface functional groups improve electrode wettability. This enhanced wettability, in turn, promotes electrolyte ion diffusion, thereby indirectly supporting the electrochemical performance [32].

Electrochemical performance

To assess the possible application value of CRK, we examined its electrochemical behavior. The CV curves at various scan rates are rectangular, indicating the presence of electric double-layer capacitance for all samples. As illustrated in Fig. 11a, CRK700's CV curve exhibits a larger area compared to CS700, CRK600, and CRK800, demonstrating that CRK700 has the highest specific capacitance as anticipated. Figure 11b illustrates that the CV curve of the CRK700 retains a more rectangular shape than that of CRK800, suggesting that the CRK700 exhibits outstanding dynamic reversibility and excellent rate performance. BET analysis reveals a critical difference in pore structure: while CRK800 has the highest surface area, it is dominated by micropores. In contrast, CRK700 features a hierarchical porous network with a substantial mesopore fraction. Consequently, the excessive microporosity in CRK800 creates significant ion diffusion limitations during CV measurements, preventing electrolyte ions from efficiently accessing the high surface area and leading to a non-ideal, diffusion-controlled CV shape.

GCD tests further highlight the superior electrochemical performance of CRK700. As shown in Fig. 11c, the triangular GCD curves exhibit the characteristic features of electric double-layer capacitance (EDLC) for all samples. The linearity of the GCD curves indicates that all samples demonstrate good ionic responsiveness in the 3 M KOH electrolyte [33]. At a current density of 1 A g^{-1}, the specific capacitances derived from Eq. 4 are 199.7, 263.27, 318, and 174.35 F g^{-1} for CS700, CRK600, CRK700, and CRK800, respectively. The excellent performance of CRK700 can be ascribed to the combined effects of its highly developed porous structure, large specific surface area, and Faradaic pseudocapacitance. Micropores enhance

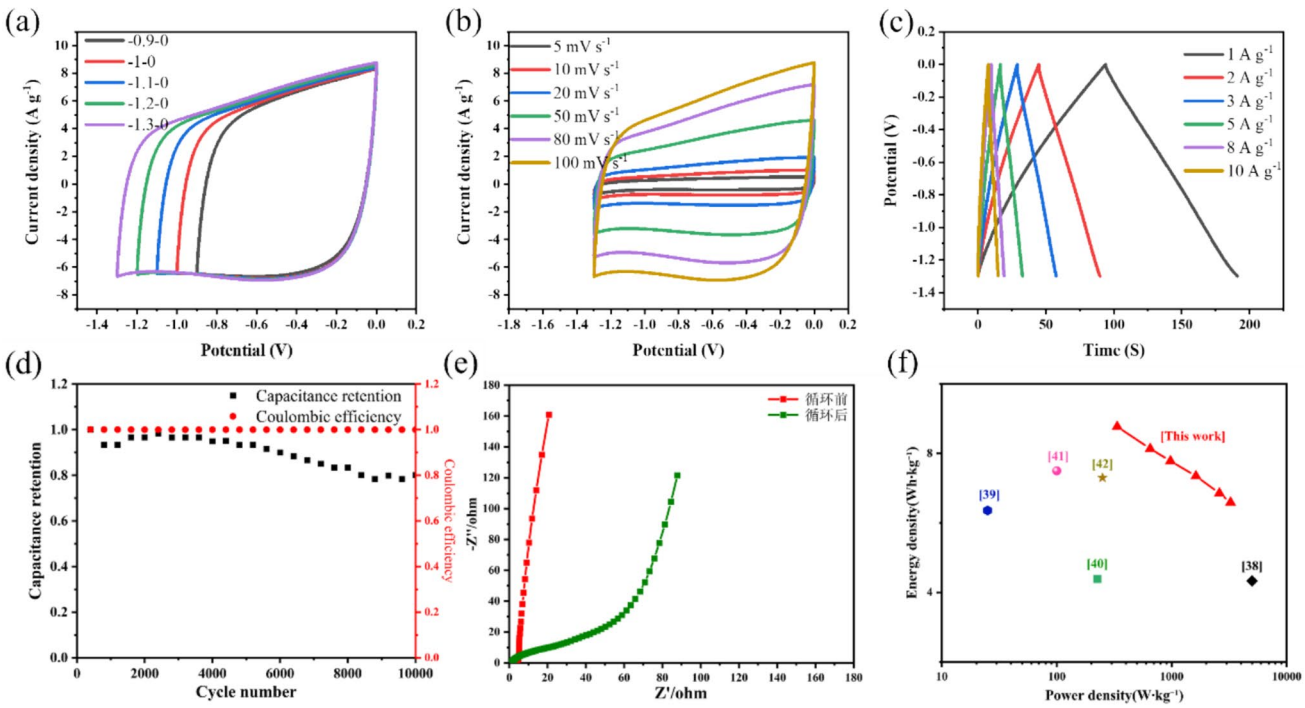

Figure 12 **a** CV curves of CRK700//CRK700 at different potential windows, **b** CV curves at different scan rates, **c** GCD plots at different current densities, **d** cycling performance and Coulombic efficiency of CRK700//CRK700 at 10 Ag^{-1}, **e** Nyquist plots before and after cycling of CRK700, **f** Ragone plots.

specific capacitance directly [34], mesopores support capacitance retention at high scan rates [35], and the redox reactions facilitated by lone-pair electrons on nitrogen atoms interacting with the electrolyte contribute to the high specific capacitance [36].

EIS test is employed to assess the charge transfer characteristics of CRK-x. In Fig. 11f, the Nyquist plots show semi-circular arcs in the high-frequency region and straight lines in the low-frequency region for all samples. For the CRK700 electrode, the low-frequency region exhibits a steeper slope compared to the other samples, indicating superior diffusion characteristics [37]. Additionally, CRK700 shows the smallest semicircle in the high-frequency region, reflecting the lowest charge transfer resistance. This suggests that the porous network structure and optimized pore size distribution of CRK700 enhance electrolyte penetration and facilitate rapid ion adsorption. Figure 11g illustrates the cycling performance of CRK700 at 10 A g^{-1}. The CRK700 electrode shows minimal capacitance degradation after 10,000 charging–discharging cycles at this current rate, indicating excellent charge–discharge cycle stability.

To further evaluate the practical application of the CRK700 electrode material, we fabricated a symmetric supercapacitor using two equal mass CRK700 electrodes in all-solid. To ensure that the potential of each individual electrode remains within its electrochemical stability range in the aqueous electrolyte during device operation, we expanded the voltage window for testing. As shown in Fig. 12a, the CV curves of CRK700//CRK700 at a scan rate of 100 mV·s^{-1}, across various voltage windows, reveal that in the −1.3 to 0 V window, the CV curve maintains a quasi-rectangular shape without significant polarization. Figure 12b shows the CV curves of the supercapacitor (CRK700//CRK700) at different scan rates. Even at a scan rate of 100 mV·s^{-1}, the curve is still quasi-rectangular, indicating that CRK700//CRK700 has excellent electrochemical double-layer capacitance performance.

Figure 12c shows the charge–discharge curves from 1 to 10 A g^{-1}, which are nearly isosceles triangles with minimal IR drop, underscoring the superior EDLC performance of CRK700//CRK700. The specific capacitance of CRK700//CRK700 at 1 A g^{-1} is calculated to be 149.49 F g^{-1}, with an energy density of 8.77 Wh kg^{-1} and a power density of 335.35 W kg^{-1} according to Eqs. 4–7. These results highlight the excellent conductivity of CRK700, surpassing other chitosan-based

supercapacitors with complex treatments [38–42], as shown in Fig. 12f.

The cycling durability of the CRK700//CRK700 is evaluated through constant-current charge and discharge tests at a current density of 10 A g^{-1}. In Fig. 12d, the specific capacitance of CRK700//CRK700 remains at 80% of the initial value after 10000 GCD cycles, indicating good cycling stability. Impedance changes before and after 10000 cycles are shown in the Nyquist plot in Fig. 12e. The equivalent series resistance (R_s) remained around 0.9 Ω, indicating its well-done material stability. However, it can be clearly observed that the charge transfer resistance (R_{ct}) increases after cycling, rising from 1.93 to 20.7 Ω. This evolution can be attributed to alterations at the electrode–electrolyte interface during cycling, potentially involving the rearrangement/loss of surface functional groups and the formation of a solid electrolyte interphase (SEI)-like film. These factors collectively increase the charge transfer resistance, leading to the observed significant changes in the EIS spectra.

In summary, CRK700 has excellent electrochemical qualities, including ultra-high specific capacitance, high multiplicity performance, and extraordinary stability. Therefore, multi-component doped porous carbon prepared from chitosan adsorbed with Congo Red has significant potential for application.

Conclusion

In this work, chitosan demonstrated a maximum adsorption capacity of 1450 mg g^{-1} for Congo Red, highlighting its effectiveness as a natural adsorbent. High-temperature carbonization of the adsorbed CS/CR complexes yielded porous carbon materials richly doped with abundant N, O, and S heteroatoms, with a specific surface area of up to 2152.12 m^2 g^{-1}. The electrode material has a specific capacitance of 318 F g^{-1} at a current density of 1 A g^{-1}. The assembled supercapacitor exhibited a specific capacitance of 149.49 F g^{-1}, an energy density of 8.77 Wh kg^{-1}, and a power density of 335.35 W kg^{-1} at 1 A g^{-1}. This strategy converts spent adsorbents into high-performance carbon materials, thereby not only tackles the challenges of adsorbent recovery and reprocessing in conventional wastewater treatment but also promotes the green recycling of materials, thereby improving resource efficiency. It offers a novel approach to integrating environmental protection with energy technologies.

Author contribution

Yang Gao contributed to conceptualization, experimentation, data curation, formal analysis, methodology, investigation, visualizations, writing—original draft, and writing—review and editing. Shan Gao helped in methodology, and supervision. yunhua zhang was involved in supervision and validation. Shaoyi Lyu carried out data curation and investigation. Feijun Wang contributed to conceptualization, resources, supervision, validation, writing—original draft, and writing—review and editing.

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Declarations

Conflict of interest We declare that there is no conflict of interesting including competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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